/EST BENGAL STATE ELECTRICITY DISTRIBUTION COMPANY LIMITED

(A Govt. of West Bengal Enterprise)

Office of the Chief Engineer,

Procurement & Contracts Department.

Vidyut Bhavan, 4th. floor,

Bidhannagar, Kolkata – 700091.

TECHNICAL SPECIFICATION

FOR

X-LINKED POLYETHYLENE INSULATED AND PVC SHEATHED H.T. (11KV) AERIAL BUNCHED CABLES FOR EFFECTIVELY EARTHED SYSTEM

TECHNICAL SPECIFICATIONS FOR X-LINKED POLYETHYLENE INSULATED AND SHEATHED H.T. (11KV) AERIAL BUNCHED CABLES FOR EFFECTIVELY EARTHED SYSTEM

) SCOPE:

This specification covers design, manufacture, testing at manufacturer's works before despatch, transportation, supply and delivery of 11 KV grade XLPE insulated, screened all Aluminium conductor and PVC sheathed Power Cables, manufactured with dry cure technology with CCV system HT Aerial Bunched Cables for overhead lines.

0 LOCATION

11KV Aerial bunched Cables are composite Cable especially intended for overhead line use in forest areas where tree clearance poses a serious problem and in other cases where it is difficult to get clearance for bare conductor lines. The said cable will be used within the State of West Bengal under WBSEDCL area.

.0 CLIMATIC AND ISOCERAUNIC CONDITIONS:

Maximum ambient temperature 50°C

Minimum ambient temperature - 4°C

Thermal resistance of soil - 150°C - cm/watt

Maximum daily average ambient temperature: 45°C

Maximum relative humidity- 100%

Average rainfall per annum-200 cm

Maximum height above the sea level -1000 meter

Maximum wind pressure/wind speed: 150 kg/sq. m/50m/sec

Average no. of thunderstorm days per annum: 200

4.0 COMPOSITION & DESCRIPTION OF THE CABLE:

The composite cable shall comprise three single-core Power cables twisted around Aluminium Alloy a bare messenger wire, which will carry the weight of cable.

Single-core Power cables should consists of **stranded compacted Circular** Aluminium Conductor screened with Black extruded semi-conducting Compound, natural coloured XLPE insulated core screened with Black extruded semi-conducting compound and one layer of copper tape and covered with Black extruded PVC.

Bare Messenger Conductor shall be stranded **non-compacted Circular** aluminium alloy conductor & will carry the weight of the cable. It shall have smooth surface to avoid damage to the outer insulating sheath of single core phase cables twisted around the messenger.

Aluminium Alloy wires shall be tested as per IS: 398(Part-IV)/1994.

5.0 DESCRIPTION OF THE CABLE

Description	Sl. No.	Designation
11KV (E) Grade Aerial Bunched Cable with 3(three) Power Cores (Stranded, compacted Circular Aluminium Conductor screened with Black extruded semi-conducting Compound, natural coloured XLPE insulated core screened with Black extruded semi-conducting compound and one layer of copper tape and covered with Black extruded PVC (Core identification by colour tape marking red, yellow & blue with colour tape placed longitudinally under the copper tape)) and one bare Messenger wire shall be Stranded non-compacted circular Aluminium Alloy conductor. The Cable should conform to IS: 8130-1984, IS:398(Part-IV)-1994 & IS:7098(Part-II)-2011 with up to date amendments, if any.	1.	3x35+1x70 sq. mm.
	2.	3x50+1x70 sq. mm.
	3.	3x70+1x70 sq. mm.
	4.	3x95+1x70 sq. mm.
	5.	3x120+1x125sq.mm.
	6.	3x150+1x148 sq mm.

EM DETAILS:

Voltage Grade of (KV) of cable required	: 6.34/11
Service Voltage	: 11kV
Highest Voltage	: 12kV
Earthing System	: Solidly Earthed
B.I.L. for Cable	: 75kV for 11kV Grade
Fault Level (Maxm.)	: See Clause 9.0
Frequency	: 50C/S

7.0 APPLICABLE STANDARDS:

Unless otherwise stipulated in this specification, the following standard with **latest amendment** shall be applicable:

IS:7098 (Part-II)-2011 and its amendment	Specification for Cross linked polyethylene insulated thermoplastic sheathed Cables for working voltages from 3.3 KV up to and including 33 KV.
IS : 8130-1984 and its amendment	Specification for Conductors for Insulated Electric Cables and flexible Cords.
IS : 398 (Part-IV)-1994 and its amendment	Specification for Aluminium Conductors for overhead Transmission purposes: Aluminium Alloy Stranded Conductors (Aluminium-Magnesium- Silicon type)
IS:10418 1982 and its amendment	Specification for Drums for Electric Cables

8.0 GENERAL TECHNICAL REQUIREMENTS:

8.1 DETAILS OF SINGLE-CORE POWER CABLE

The cable conductors shall be H4 grade class 2 Aluminium Conductors complying with the requirements as specified in IS-8130-1984 **with latest amendments.**

The conductor shall be stranded compacted circular. Conductor shall be clean & reasonably uniform in size and shape and its surface shall be free from sharp edges.

Not more than two joints shall be allowed in any of the wires forming every complete length of conductor and no joint shall be within 300 mm. of any other joint in the same layer. Joints shall be brassed, silver soldered or electric or gas welded. No joint shall be made in the conductor, once it has been stranded.

Three Power Cores should be identified by colour tape marking (Red, Yellow & blue) with colour tape placed longitudinally under the copper tape

8.2 Conductor Screen

The conductor screen shall be black extruded semi-conducting compound of thickness **0.5 mm. (minimum).**

8.3 Insulation of XLPE

The insulation shall be of cross linked polyethylene (XLPE) of nominal insulation thickness 3.6 mm. conforming to the physical, electrical and ageing properties of IS: 7098 (Part-II):2011 with latest amendments therof.

Cross-linking should be done by exposure to peroxide with **Nitrogen Curing CCV Line only** method with intention to ensure lowest tree formation. Only natural unfilled compound shall be used for insulation.

XLPE insulation shall be done by method of extrusion. The insulation shall be so applied that it fits on the conductor (or conductor screening or barrier, if any) and it shall be possible to remove it w. damaging the conductor.

The XLPE conductor shall be suitable for use where the combination of ambient temperature and temperature rise due to load, including temperature on exposure to direct sunlight results in conductor temperature not exceeding the following:

Normal continuous Operation	Short circuit Operation
90^{0}C	250^{0}C

.4 Insulation Screen

The insulation screening shall consist of two parts; namely non metallic & metallic.

Non metallic part shall consist of a layer of extruded semi-conducting compound and shall be applied directly over the insulation of each core. Thickness of non-metallic part should be **minimum 0.6 mm.**

The metallic part shall be non magnetic and shall be applied over the non-metallic part of insulation screening of individual core. Metallic screening material shall be annealed Copper tape. Thickness of Copper tape should be **minimum 0.06 mm; width of the copper tape should be minimum 50 mm and overlapping of copper tape should be minimum 15%.**

A layer of non-conducting water swell-able tape (0.3 mm thickness) applied helically with minimum 50% overlapping over the Copper tape.

8.5 Outer Sheath

Over the metallic screen the Cable shall be provided with extruded PVC outer sheath. The composition of the PVC compound for outer sheath shall be Type ST2 of IS 5831-1984 with latest amendments.. The colour of the outer sheath shall be black. The nominal & minimum thickness of the sheath shall not be less than the standard values specified in the table No. 7 of IS-7098 (Part-II)/2011 with up to date amendments, if any. Outer surface of the Cable shall be ultra-violet Ray resistant.

9.0 Dimensional And Electrical Data

The standard sizes and technical characteristics for Single Core Power Cable is given below:

Nominal Cross sectional Area of Al. Conductor (sq. mm)	Maximum DC resistance at 20^{0}C ohm/km	Minimum no. of Strand	Nominal thickness of insulation (mm)	Max. Short Circuit Current for 1 sec. (KA)
35	0.868	6	3.6	3.4
50	0.641	6	3.6	4.72
70	0.443	12	3.6	6.7
95	**0.32**	**15**	**3.6**	**8.96**
120	0.253	15	3.6	11.32
150	0.206	15	3.6	14.16

Note : The resistance values given in the Table are the max. permissible one.

10.0 DETAILS OF MESSENGER (NEUTRAL CONDUCTOR)

Bare Messenger Conductor shall consist of aluminium alloy wires. Aluminium Alloy Wires shall be tested as per IS:398 (Part-IV) /1994 **with latest amendments,** if any.

Messenger Conductor shall be Stranded non-compacted Circular & shall have smooth round surface to avoid damage to the outer insulating sheath of single core phase cables twisted around the messenger.

There shall be no joints in any wire of the stranded messenger conductor except those made in the base rod or wires before finally drawing. The technical characteristics of the messenger wire shall be as follows:

Nominal Cross sectional Area of Phase Conductor (sq. mm)	Messenger Conductor (Stranded Al Alloy IS 398 Part-4) Nominal Area of Aluminium Alloy in Messenger Wire (sq. mm)	No. of Strand of Aluminium Alloy	Strand dia. of Aluminium Alloy (mm)	Maximum DC resistance at 20°C of messenger Conductor (ohm/km)	Minimum breaking load of Messenger Conductor (KN)
35	70	7	3.57	0.492	20.458 (approx)
50	70	7	3.57	0.492	20.458 (approx)
70	70	7	3.57	0.492	20.458 (approx)
95	**70**	**7**	**3.57**	**0.492**	20.458 (approx)
120	125	19	2.89	0.27	36.64
150	148	19	3.15	0.23	43.50

11.0 LAYING UP:

Three Power Cores **having colour tape marking** shall be twisted over bare messenger wire with right hand direction of lay. This will from the Aerial Bunched Cable.

12.0 TESTS:

All the cable sizes i.e. items offered should have been fully type tested as per the relevant standards at any NABL accredited Lab bearing an Accreditation body Logo/Central or State Govt. recognised test house or laboratory.

The bidder shall furnish the type test reports along with the offer. These type tests must have been conducted within last five years prior to date of Bid opening.

Valid Calibration Certificate of instruments/equipment used for Testing purpose conducted by NABL accredited Laboratory provided the certificate bears an NABL accreditation body logo. For testing equipment where NABL accreditation is not available, calibration certificate from educational institutions like IIT's, NIT's, J.U., C.U., B.H.U. only can be accepted provided they demonstrate traceability.

The following Type tests shall be carried out on the cables as per Relevant IS. & Type tests certificates shall be furnished invariably with the offer.

LIST OF TYPE TESTS :

a) i) Resistance Test on Conductor.

b) Test for thickness of insulation and sheath.

c) **Physical tests for insulation.**

i) Tensile strength and elongation at break.

ii) Ageing in air oven.

iii) Hot set test.

iv) Shrinkage test.

v) Water absorption (Gravimetric)

d) **Physical Test for outer sheath :**

i) Tensile Strength and elongation at break.

ii) Ageing in on oven.

iii) Shrinkage Test.

iv) Hot deformation.

v) Loss of mass in air oven, heat shock and thermal stability Test

e) Partial discharge test.

f) Bending test.

g) Dielectric Power Factor Test :

i) As a function of Voltage.

ii) As a function of temperature.

h) Insulation resistance (volume resistivity) test.

i) Heat cycle test.

j) Impulse withstand test.

k) High Voltage test.

l) Flammability test.

m) Cold impact test

n) **Type test for messenger wire :**

i) Ultimate Tensile Strength test on finished wire

ii) Resistance Test

ACCEPTANCE TESTS :

The following tests shall be carried out as acceptance tests :

a) Conductor resistance test for both messenger and Phase Conductor.

b) Test for thickness of insulation and sheath of Phase Conductor.

c) Partial discharge test (for screened cables only)

d) High Voltage test.

e) Hot set test for Insulation

f) Tensile Strength & elongation at break test for insulation & Sheath

g) Insulation resistance (volume resistivity) test

Nominal Cross sectional Area of Phase Conductor (sq. mm)	Messenger Conductor (Stranded Al Alloy IS 398 Part-4) Nominal Area of Aluminium Alloy in Messenger Wire (sq. mm)	No. of Strand of Aluminium Alloy	Strand dia. of Aluminium Alloy (mm)	Maximum DC resistance at 20°C of messenger Conductor (ohm/km)	Minimum breaking load of Messenger Conductor (KN)
35	70	7	3.57	0.492	20.458 (approx)
50	70	7	3.57	0.492	20.458 (approx)
70	70	7	3.57	0.492	20.458 (approx)
95	**70**	**7**	**3.57**	**0.492**	20.458 (approx)
120	125	19	2.89	0.27	36.64
150	148	19	3.15	0.23	43.50

11.0 LAYING UP:

Three Power Cores **having colour tape marking** shall be twisted over bare messenger wire with right hand direction of lay. This will from the Aerial Bunched Cable.

12.0 TESTS:

All the cable sizes i.e. items offered should have been fully type tested as per the relevant standards at any NABL accredited Lab bearing an Accreditation body Logo/Central or State Govt. recognised test house or laboratory.

The bidder shall furnish the type test reports along with the offer. These type tests must have been conducted within last five years prior to date of Bid opening.

Valid Calibration Certificate of instruments/equipment used for Testing purpose conducted by NABL accredited Laboratory provided the certificate bears an NABL accreditation body logo. For testing equipment where NABL accreditation is not available, calibration certificate from educational institutions like IIT's, NIT's, J.U., C.U., B.H.U. only can be accepted provided they demonstrate traceability.

The following Type tests shall be carried out on the cables as per Relevant IS. & Type tests certificates shall be furnished invariably with the offer.

LIST OF TYPE TESTS :

a) i) Resistance Test on Conductor.

b) Test for thickness of insulation and sheath.

c) **Physical tests for insulation.**

- i) Tensile strength and elongation at break.
- ii) Ageing in air oven.
- iii) Hot set test.
- iv) Shrinkage test.
- v) Water absorption (Gravimetric)

d) **Physical Test for outer sheath :**

i) Tensile Strength and elongation at break.

ii) Ageing in on oven.

iii) Shrinkage Test.

iv) Hot deformation.

v) Loss of mass in air oven, heat shock and thermal stability Test

e) Partial discharge test.

f) Bending test.

g) Dielectric Power Factor Test :

i) As a function of Voltage.

ii) As a function of temperature.

h) Insulation resistance (volume resistivity) test.

i) Heat cycle test.

j) Impulse withstand test.

k) High Voltage test.

l) Flammability test.

m) Cold impact test

n) **Type test for messenger wire :**

i) Ultimate Tensile Strength test on finished wire

ii) Resistance Test

ACEPTANCE TESTS :

The following tests shall be carried out as acceptance tests :

a) Conductor resistance test for both messenger and Phase Conductor.

b) Test for thickness of insulation and sheath of Phase Conductor.

c) Partial discharge test (for screened cables only)

d) High Voltage test.

e) Hot set test for Insulation

f) Tensile Strength & elongation at break test for insulation & Sheath

g) Insulation resistance (volume resistivity) test

Breaking load test on finished wire for Messenger Wire

Resistance Test for Messenger Wire

j) Flame retardance test on bunched cables

ROUTINE TEST ;

The following tests shall be carried out as routine tests on all sizes of all drums of un armoured cables by the supplier in presence of the purchaser's representative if so directed by the purchaser and shall be got approved before despatch.

a) Conductor resistance test (for aluminium)

b) Partial discharge test (for screened cables only)

c) High Voltage Test.

Type test/acceptance test/Routine tests and any other tests not covered as stated in different clauses of the specification but required as per relevant Indian Standards shall also to be carried out.

13. INSPECTION :

The Cable shall be inspected at manufacturer's works before despatch as per IS-7098 (Part- II)/2011 **with latest amendments** thereof. All the acceptance tests embodied in the above shall be performed by the Inspecting Officer. The Manufacturer shall arrange without making any extra charges with all the necessary machinery, apparatus and labour requirement for the testing purpose. The cost requirement for testing shall be to Firm's Account.

6 copies of type test certificate lot wise for each type of cable shall be sent to the purchaser for acceptance. Type test certificate for each lot and routine test certificate for each drum of cables shall be submitted to purchaser for approval before despatch of cables from the works. The test certificate shall be completed with all results.

14. CABLE IDENTIFICATION

The following shall be embossed on the outer sheath for the identification.

a) Manufacturer's Name or Trade Name/Trade Mark.
b) Type of Cable i.e. HT AB Cable
c) Voltage Grade (11KV).
d) Type of insulation i.e. XLPE, material of conductor i.e. Aluminium.
e) Nominal cross-sectional area of the Power Conductors & Bare messenger conductor.
f) Month & Year of manufacture.
g) Inscription for length of cables at 1.0 meter interval on any one of phase cores by printing/ engraving.
h) Name of the purchaser : WBSEDCL/DDUGJY/IPDS (as applicable)
i) Marking "Electric-WBSEDCL" shall be embossed throughout the length of the Cable at 1.0 metre spacing on any one phase conductor.

15. SEQUENTIAL MARKING :

Supplier shall provide non-erasable sequential marking of length on any one of phase cores by printing/engraving for each metre of length with sequential marking machine having automatic length measuring meter. The Cable supplied without sequential marking will not be accepted and no deviation shall be allowed under any circumstances.

16. PACKING & MARKING :

The Cable shall be supplied in suitable non-returnable wooden drums of suitable size in standard lengths subject to a tolerance of ± 5%. Manufacture will provide the Multiplying Factor considering the Lay Ratio of the cable to measure the length of cable in a drum. Multiplying factor would be provided in the GTP instead of drum. Sequential length of the core & the declared cable length shall be marked on the cable drum. Generally IS: 7098-II/2011 **with latest amendments thereof** shall be marked on the cable drum.

The Cable shall be wrapped with polythene under wooden covering. Each metre length shall be embossed with the Trade Name of Manufacturer and the word 'ELECTRIC WBSEDCL'. Drums shall be proofed against attack by white ants or termite conforming to IS: 10418The ends of the Cable shall be sealed by means of non-hygroscopic sealing material. Each drum shall carry the following information either stencilled on flange or label attached to it

a) Manufacturer's Name & Trade Mark, if any
b) Drum Identification no..
c) A reference to ISS i.e. IS-7098 (Part-II)/2011.
d) Length of the Cable on the drum.
e) Voltage Grade and type of the Cable.
f) Number of Cores (Power & Messenger).
g) Nominal cross-sectional area of the Cable Conductor.
h) Purchase Order No. & date.
i) Name and address of Purchaser – WBSEDCL or Project Name (whichever is applicable).
j) Colour of outer sheath.
k) Gross Weight of packed and net weight of the composite Cable.
l) Drum No. & its direction of rotation (with an arrow)
m) Month & Year of Manufacture.
n) Month of delivery

The standard Drum Length will be 250 meters. in each Drum for all sizes subject to a maximum tolerance of **± 5%. The overall tolerance limit of total delivered quantity should be minus (-) 1%** against item wise ordered quantity for each Purchase Order and the same shall be taken care of while offering the last lot of inspection.

17. LITERATURE AND MANUAL :

To be submitted as per General Terms & Conditions of Contract. Following documents are to be submitted at the time of physical delivery at consignee stores

i. Copy of Purchase Order
ii. Copy of dispatch instruction
iii. Inspection Test certificate
iv. Guarantee certificate
v. Proforma Invoice
vi. Calculation Sheet for price variation on the basis of IEEMA with base date of order
vii. Seal list and packing list
viii. Challan in triplicate
ix. e-Way Bill, if applicable

18. GUARANTEED TECHNICAL PARTICULARS :

To be submitted along with Tender documents in triplicate as per format annexed herewith Technical characteristic shall be guaranteed by the bidder. In case of failure of materials to meet the guarantee, WBSEDCL shall have right to reject the material.

GUARANTEED TECHNICAL PARTICULARS
OF
11 KV HT AB CABLE

(To be filled and signed in by the Supplier)

1	Manufacturer's Name & Address	
2	AB Cable Sizes	3X ....+1X .... sq. mm
3	Lists of Standard applicable.	IS: 7098 (Part II)-2011, IS: 8130-1984, IS: 398 (Part-IV) 1994, IS: 10418: 1982.
4	**Details of Power Core Conductor**	
a)	Material	Aluminium conductor shall be H4 grade complying with the requirements of IS-8130-1984 with up to date amendments
b)	Flexibility Class as per IS : 8130-1984	Class 2
c)	Form of Conductor.	Stranded Compacted Circular
d)	Maximum Continuous Conductor Temperature (in ^{0}C)	90^{0}C
e)	Maximum Short time Conductor Temperature (in ^{0}C)	250^{0}C
5	**Power Core**	
a)	No of Core	3
b)	Nominal cross sectional area (sq.mm.)	
c)	No. of strands	
d)	Strand dia. (mm.)	
e)	Dia of compacted Conductor (mm.)	
f)	Insulation Nominal Thickness (mm.)	
g)	Insulation Minimum Thickness(mm.)	
h)	Approximate dia over insulation (mm.)	
i)	Maximum Continuous Load (Amps)	
j)	Maximum Short Circuit current for 1 sec (KA)	
k)	Identification of Power Cores	By colour Tape marking (Red, Yellow & blue)
l)	Max. DC resistance of conductor at 20^{0}C(ohm/km)	
m)	Approx. Mass (Kg/KM)	
6	**Bare Messenger Core**	
a)	Material	Aluminium Alloy Wires.
b)	No of Core	1
c)	Nominal cross sectional area (sq mm.)	
d)	No. of strands	
e)	Strand dia (mm.)	
f)	Dia of non compacted Conductor (mm.)	
g)	Approx. Weight (Kg/KM)	
h)	Max. DC resistance of conductor at 20^{0}C (ohm/Km)	

i)	Ultimate tensile Strength (Kg)	
j)	Earth fault current carrying capacity for 1 sec (KA)	
k)	Modulus of Elasticity (Kg/cm²X10⁵)	
l)	Co-efficient of linear expansion (per ⁰CX 10⁻⁶)	
7	**Insulation**	
a)	Material	XLPE
b)	Method of application of Insulation	Extrusion
c)	Type of curing of XLPE Insulation Completed Cable.	Nitrogen Curing through CCV line only
d)	Colour of Insulation	Natural Coloured
8	**Conductor Screening**	
a)	Material	Semi conducting XLPE Compound
b)	Minimum Thickness	0.5 mm
c)	Method of application	Extrusion
9	**Insulation Screening**	
A	**Non-Metallic Part**	
a)	Material	Semi conducting compound
b)	Minimum Thickness	0.6 mm
c)	Method of application	Extrusion
B	**Metallic Part**	
a)	Material	Annealed Copper Tape
b)	Minimum Thickness	0.06 mm
c)	Minimum Width	50mm
d)	Minimum Overlapping	15%
e)	Non-Conducting Water Swell-able tape on Copper Tape	0.3mm Nominal with minimum 50% overlapping
10	**Outer Sheath**	
a)	Material	PVC compound of type ST2 of IS 5831-1984
b)	Nominal Thickness (mm.)	
c)	Method of application	Extrusion
d)	Colour	Black
11	**Voltage Grade of Cable**	
a)	Maximum operating voltage of the Cable	12 KV
b)	Rated Voltage of the Cable	11 KV
c)	Neutral to Phase voltage	6.35 KV
12	De-rating factor: De-rating factors for variation in Air Temp.	
	Air Temp. (⁰C)	
13	**Direction of Laying**	Right Hand Direction
14	**Completed Cable :**	
a)	Approx. overall diameter(mm).	

	Approx. Weight(Kg/KM)		
	Allowable sag as a percentage of span length at 4^{0}C ambient temperature		
15	**Cable Drums :**		
a)	Standard drum length & tolerance of each drum	250 Mtr, ± 5%	
b)	Dimension of Drum	As per IS:10418-1982	
c)	Approx. Shipping Weight. (KG/ Drum))		
16	**Overall quantity tolerance limit**	-1%	
17	**Sequential marking**	Marking of the sequential length shall be provided on any one of phase cores by printing/ engraving for each meter of length.	
18	**Embossing**	Each meter length of cable shall be embossed with the following on any one phase conductor. a) Manufacturer's Name or Trade Name/Trade Mark. b) Type of Cable i.e. HT AB Cable c) Voltage Grade (11KV). d) Type of insulation i.e. XLPE, material of conductor i.e. Aluminium. e) Nominal cross-sectional area of the Power Conductors & Bare messenger conductor f) Month & Year of manufacture. g) Inscription for length of cables at 1.0 meter interval on any one of phase cores by printing/ engraving. h) Name of the purchaser : WBSEDCL/DDUGJY/IPDS (as applicable) i) Marking "Electric WBSEDCL" shall be embossed throughout the length of the Cable at 1.0 metre spacing on any one phase conductor.	
19	**Marking on each drum**	The following information shall be stenciled on each drum: a) Manufacturer's Name, Trade Name/Trade Mark. b) Drum identification No. c) IS reference i.e 7098 (part II)/2011 d) Length of the Cable in Cable Drum. e) Type of Cable and Voltage Grade f) No. of Cores(power & messenger). g) Nominal Cross sectional area of the conductor (power & messenger) of the cable. h) Colour of outer sheath i) Direction of rotation of Drum (with an arrow) j) Gross Weight of packed and net weight of the composite cable k) Month & Year and Country of Manufacture l) Purchase Order No. & date m) Month of Delivery n) Name of the Purchaser : WBSEDCL/Project Name (as applicable)	
20	**Multiplying Factor considering Lay Ratio of cable to Measure the Length of cable in Drum**		

Signature with designation & Seal
With Name of the Firm